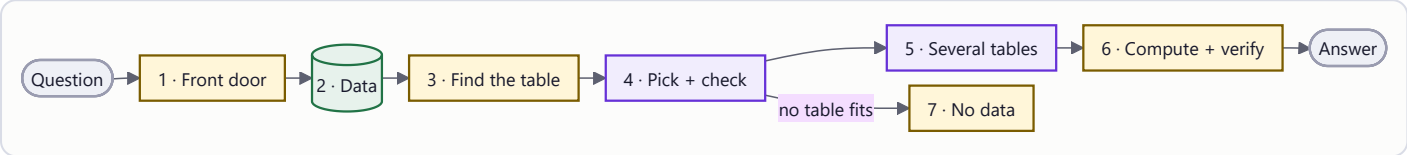
Spurðu Hagstofuna (Ask Statistics Iceland)

Project description and results · Gagnaþon 2026, Track 2 (direct PX-API) · Team: Atli and Soukaïna, who built it together · Code: github.com/Atlimh1/Tjarna

What we set out to achieve

A chatbot anyone can ask about Statistics Iceland's figures, in Icelandic, in English or in another language, that answers with the official number, its unit and period, and the table it came from, and that never invents a number. One principle runs through the design: **the model chooses, code does the rest**. The language model picks the table and the values and writes the sentence; code finds the candidate tables, reads the numbers, does every calculation, checks every number in the answer against the data and cites the source, with the exact PX-API query that reproduces it.

How it works

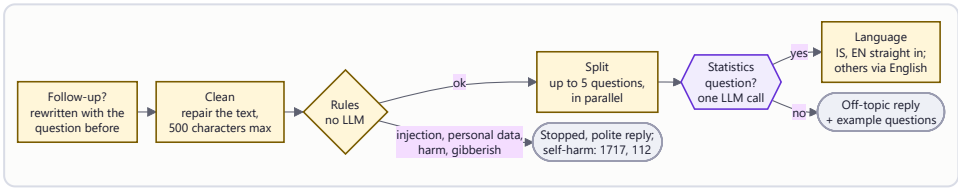


Architecture overview; each numbered part has its own diagram below.

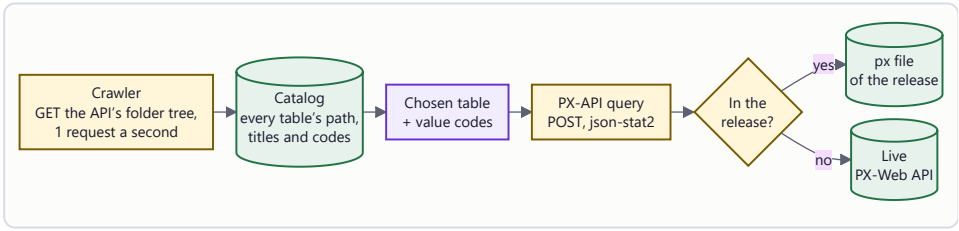
language model (Gemma 4 31B) code data question and answer

Runs on: vLLM serving Gemma 4 31B on 4 A100 GPUs on the JURECA supercomputer (Jülich), where our search model was also fine-tuned; the pipeline, a FastAPI service and the web page on a Raspberry Pi 5 or a JURECA login node.

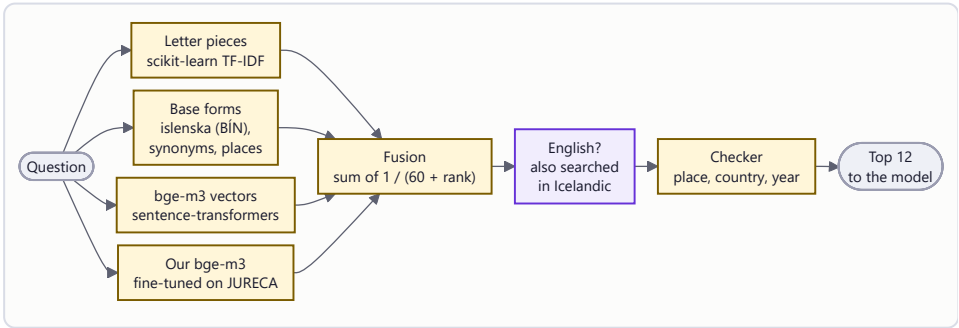
1 · Front door: many rule checks, then one last LLM check. Rules with no model stop injection, personal data, harm, abuse, gibberish and plain sums. One LLM call is the last check: is it a statistics question at all? Our table picker always picks a table, so without it an off-topic message would get a number from some table; the rules stopped 0 of 16 off-topic inputs, and a search-score cut-off could not tell them apart.

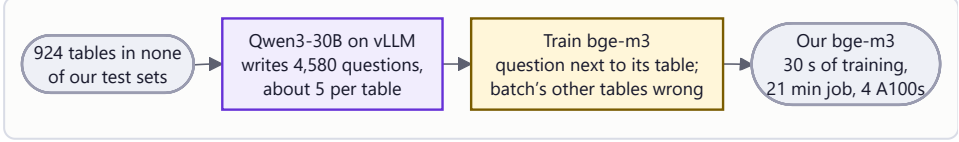


2 · The data (Track 2). The organizers' release, 1,886 tables in Icelandic and English as of 18 Oct 2026, and Statistics Iceland's public PX-Web API. No question comes with its table. Our crawler walked the API's folder tree, one request a second, and saved every table's path, titles and value codes; every answer shows its exact PX-API query, and a button re-reads it live.

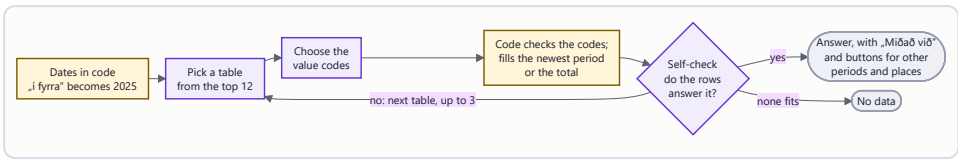


3 · Finding the table, with no language model (the starter kit's weak spot). Four searches over one index of both languages, joined by reciprocal rank fusion; table vectors are computed once and compared by cosine similarity. **How we fine-tuned bge-m3**, in one JURECA job of 21 minutes on 4 A100s: Qwen3-30B, served with vLLM, wrote about five questions per table, 4,580 for 924 tables that are in none of our test sets; then bge-m3 learned to place each question next to its own table, with the other tables in its batch of 32 as wrong answers (contrastive training, one pass, 144 steps, 30 seconds). With it the full system got 56 of the organizers' 60, against 54 with the base model.

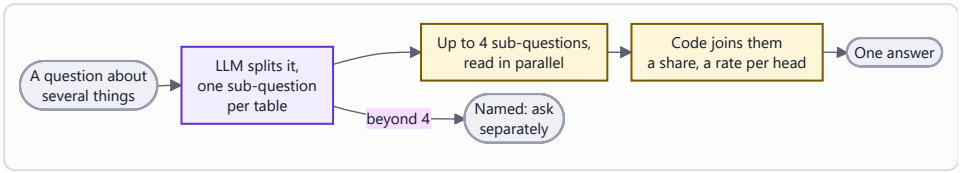




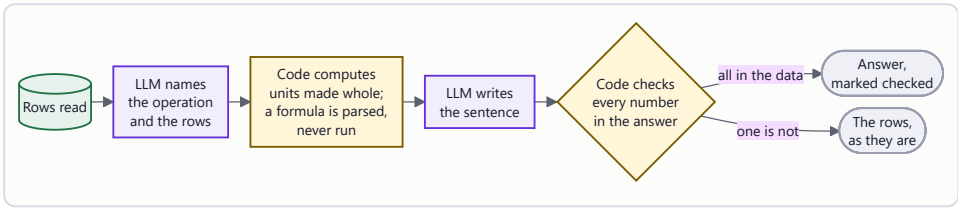
4 · The model chooses, code checks. The model picks the table and the value codes; code drops codes the table does not have, fills what the question left open with the newest period or the total, says so ("Miðað við: Ár 2026") and offers other periods or places as buttons. A year not published yet gets the newest figure, said so; a look-alike table that holds another figure is named under the answer.



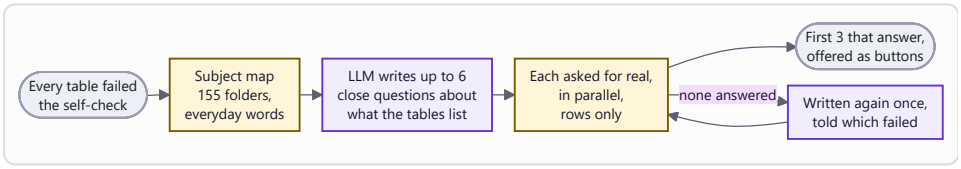
5 · Several tables. The model splits a question into up to four sub-questions, one table each, read in parallel; code can join them with a calculation; anything beyond four is named in the answer, never dropped.



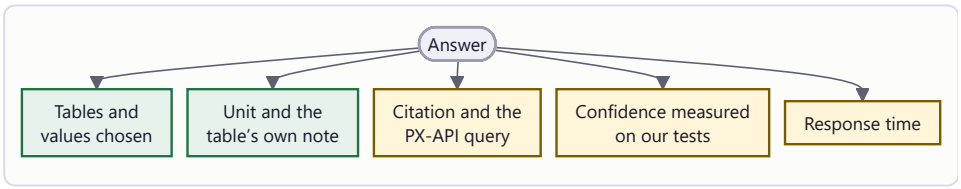
6 · Code computes and checks. The model only names the operation: differences, percent changes, shares, ratios, rates per 1,000, sums, averages and whitelisted multi-step formulas. Code computes it, with units made whole first (millions of krónur per inhabitant), and checks every number in the written answer against the data.



7 · No data, said plainly. A subject map (the release's 155 folders, each with everyday words from one LLM call, 148 words corrected by hand) adds the subject's tables. "How many apples are eaten in Iceland?" has no table; it offers potatoes, tomatoes and cereal grain in 2025, each one answered.



8 · Every answer shows what is needed to check it or reproduce it.



Results on Statistics Iceland's data

Benchmarked with the organizers' own `gagnvist evaluate`, unchanged: an answer is right only if it states every expected number and cites a table that holds it. Expected numbers are read from the data release by code, not typed.

Question set	What it is	Right
Organizers' 60	their test questions: easy, medium (calculation), hard (two tables); Icelandic and English	**60 / 60**
Set 1 (frozen)	41 new questions we wrote before any fix aimed at them	**39 / 41**
Set 2 (frozen)	45 questions written by a separate AI agent from the catalog; never tuned on	**37 / 45**
real2 (frozen)	42 questions typed by us the way people type: typos, five languages, traps	**33 / 42**
dev3	62 questions over 61 tables no other set uses, written blind on 1 Nov (typos, relative dates, places)	**45 / 62**
Should be declined	questions Statistics Iceland does not publish (earthquakes, a firm's profit, forecasts)	**27 / 28**
Other languages	19 languages, 2 questions each: the same numbers as the English original	**37 / 38**
Guard	bad inputs stopped (injection, gibberish, harm, personal data, off-topic); real questions wrongly refused	**165 / 165; 2 of 1,008**

What each step adds (the organizers' 60, Gemma 4 31B; each rung adds one part of the system):

Model alone	Embedding search, like the starter kit	Our search	+ picking the table	+ calculation in code	+ sub-questions	Full system
0	**7**	**16**	**22**	**38**	**58**	**60**

Baselines and speed. The starter kit answered 7 of its own 52 test questions, and the organizers' pipeline run with our Qwen3-30B 8 of the 60. Ours takes about 4 to 5 s per answer; 10 questions sent at once were all answered in 19.4 s, with no errors. The 60 shaped many of our rules, so the frozen and blind sets (73% to 95%) are the fairer picture of new questions.

What we built, against the tracks

Our track is Track 2 (direct PX-API). We also did the core of Track 3 (retrieval on HPC), on the JURECA supercomputer instead of LUMI.

Asked for	What we did
Track 2: extend table search and GET/POST queries	All 1,886 tables reachable through our own four-way search (the Track 2 starter answered from three hand-picked tables); every table's metadata crawled with GET, numbers read with POST queries or from the release
Track 2: multiple questions in one API call; concurrency	Up to 10 questions per API call and 5 per chat message, answered in parallel; one question can span up to 4 tables; 10 at once in 19.4 s, no errors
Track 2: validate dimensions, filters, units and sources; include the source and selected dimensions	Value codes validated, units made whole, a self-check on the rows and every number checked against the data; every answer shows the table, the values chosen, the unit, the citation and the PX-API query
Track 2: your own frontend	A web page in three views (users, presentation, developers) and a chat, on FastAPI with the starter's own endpoint and fields
Track 2: Ollama integration	**Not done.** The model is served by vLLM through the same OpenAI-compatible API that Ollama offers, but we did not test Ollama
Track 3: embeddings and retrieval on HPC	On JURECA, not LUMI: bge-m3 table vectors computed on GPUs, bge-m3 fine-tuned on 4,580 questions (21 minutes on 4 A100s), Gemma 4 31B served with vLLM
Track 3: compare retrieval quality and performance	On the organizers' 60, an embedding search alone gets 7 right and our fused search 16 (before any model picks); with the fine-tuned bge-m3 the full system got 56 against 54 with the base one (31 Oct)

Limits

- Look-alike tables that count differently (dental care paid by households or by all private payers): the answer names the other figure, but its first choice can be the less fitting table.
- Some questions have two official figures (turnout by two definitions); years before 1998 split by place are rarely found (Reykjavík in 1950).
- The release is the snapshot of 18 Oct 2026, with a live check; the model runs as a booked job on a supercomputer, and the Raspberry Pi runs three of the four searches (no GPU there for our fine-tuned model).

What we would do next

- Test on real questions from real people, and fine-tune the search with them and with more English questions.
- Tell look-alike tables apart better: table cards with each table's notes made the picker worse when we measured them (the 60 fell from 60 to 54), so the notes are shown to the user under each answer instead.
- Run on smaller hardware (a single GPU or a DGX Spark) with a smaller model.

AI tools used

- **Claude Code** (Anthropic, Claude models) as a programming assistant under our direction: writing and refactoring code, running and analysing experiments on JURECA, building evaluation sets (set 2 and dev3 were written by separate agents that never saw the pipeline), and drafting documentation. Every change was measured on held-out question sets before it was kept, and we reviewed and decided on each design choice.
- **Claude** (claude.ai chat) to plan the work and get second opinions on our choices.
- **Models inside the product**, all open-weight and free to use: Gemma 4 31B (answers), Qwen3-30B-A3B (second opinion, and the questions our search model was fine-tuned on), bge-m3 and our fine-tuned bge-m3 (search), with GreynirCorrect and BÍN for Icelandic spelling and base forms.

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